LAB 09

Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

Item	Detail
Topic	Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca
Pipeline stage	Evaluation
TSC reference	A3, K3
Duration	25 min
Folder	labs_fullcode/Lab09-performance-metrics/
Files	Lab09-performance-metrics/evaluate_stage.py, pipeline/evaluate.py, common/metrics.py, Lab09-performance-metrics/solution/evaluate.py
Prompts	Lab09_Prompts.pdf in this folder (step -> prompt, ready to copy)
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

The investment committee wants to know which of the pipeline's signals deserves capital. Return alone cannot answer that, so the evaluation stage runs every signal through the same backtester, over the same window, with the same costs, and compares it with simply holding SPY.

Objective

Build `python -m pipeline evaluate` to compute total return, CAGR, volatility, Sharpe, Sortino, maximum drawdown, Calmar, win rate, profit factor, exposure, turnover, alpha and beta for every signal against a buy-and-hold benchmark, and interpret the ranking.

Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

How to Run the Python Scripts

Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.

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Step	Command / file
Open a terminal in the labs_fullcode folder	cd labs_fullcode
Create and activate a virtual environment (once)	python3 -m venv .venv source .venv/bin/activate (Windows: .venv\Scripts\activate)
Install the libraries (once)	pip install -r requirements.txt
Optional: your Alpaca paper keys (leave blank to run OFFLINE)	cp .env.example .env (Windows: copy .env.example .env) then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env
Run Lab 09	python3 Lab09-performance-metrics/evaluate_stage.py (same as: python -m pipeline evaluate)
The complete code	pipeline/evaluate.py (reference copy: Lab09-performance-metrics/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

Lab Steps - In Python with Your AI Assistant

AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab09-performance-metrics/Lab09_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.

Step 1: Read the metric formulas (K3)

Open common/metrics.py; each function's docstring gives its formula (annualised with 252 trading days).

AI prompt - paste into your AI coding assistant
Explain each function in labs_fullcode/common/metrics.py with its formula and one sentence on what a portfolio manager learns from it; say which metrics need trade-level returns and which need a benchmark. Do not change the file.

Check: You can explain the difference between Sharpe (all volatility) and Sortino (downside only).

Step 2: Build the evaluation stage (A3, K3)

Ask your assistant for pipeline/evaluate.py.

AI prompt - paste into your AI coding assistant
Create labs_fullcode/pipeline/evaluate.py. evaluate_strategy(bars, name, costs, start, bench_ret, **params) backtests pipeline.signals.target_position(name, bars, **params) with common.backtest.backtest (next-bar open fills, costs = pipeline.config.cost_model(cfg)), slices returns and position from `start` and returns common.metrics.summary(...) with trade net returns and the benchmark. run(symbol='SPY'): start = first date the 200-day SMA exists (so every signal is warm), benchmark = buy_and_hold sliced from start; evaluate every signal in SIGNALS; print a table (rows = strategies + buy & hold, columns CAGR, Ann. volatility, Sharpe, Sortino, Max drawdown, Calmar, Trades, Win rate, Profit factor, Exposure, Turnover, Alpha, Beta) ranked by Sharpe and formatted with common.metrics.format_table; list which strategies beat buy and hold on Sharpe and how many have a smaller drawdown; print 'Sharpe by hand' for trend_filter (mean / stdev x sqrt(252)), the worst day and SPY's drawdown trough; save _outputs/Lab09_metrics.csv and chart Lab09_metrics.png (top 4 by Sharpe plus buy & hold, equity and drawdown). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline evaluate --help shows --symbol.

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Step 3: Run the evaluation (A3)

Run: python -m pipeline evaluate --symbol SPY (same as: python3 Lab09-performance-metrics/evaluate_stage.py).

Check: Window 2018-10-16 to 2025-12-31, Alpaca-style costs. ema_cross: CAGR 11.09%, vol 11.87%, Sharpe 0.95, max drawdown -14.19%, Calmar 0.78, 26 trades, win rate 50.00%, profit factor 3.54, beta 0.35. SPY buy & hold: 15.28%, 19.93%, 0.81, -33.72%, 0.45. trend_filter: 9.70%, 12.42%, 0.81, -19.36%, 24 trades, win 41.67%, profit factor 3.79, exposure 78.53%, beta 0.38. Only ema_cross beats buy & hold on Sharpe; 7 of 8 have a smaller drawdown.

Step 4: Check one metric by hand (K3)

Read the 'Sharpe by hand' line and confirm it equals the table.

Check: mean 0.00040 / stdev 0.00782 x sqrt(252) = 0.81, the same as trend_filter's Sharpe in the table.

Step 5: Interpret trade and market metrics (A3, K3)

Explain why trend_filter can win only 41.67% of trades yet have a profit factor of 3.79, and why beta is 0.38 when exposure is 78.53%.

AI prompt - paste into your AI coding assistant
Using labs_fullcode/_outputs/Lab09_metrics.csv, write a 120-word note for Lion City Capital's investment committee comparing ema_cross, trend_filter and SPY buy & hold on return, risk and risk-adjusted return. Quote the numbers, explain win rate versus profit factor, and do not recommend live trading.

Check: Few large winners pay for many small losers; the strategy is out of the market in many stress periods, so its beta is far below its exposure.

Step 6: Locate the worst periods (K3)

Read the worst-day and drawdown-trough line and find those dates on _outputs/Lab09_metrics.png.

Check: Worst trend_filter day 2020-06-11 (-5.76%); worst SPY day 2020-03-16 (-10.94%); SPY drawdown trough 2020-03-23.

Step 7: Add a metric to the stage (K3)

Extend the evaluation table with one more risk metric.

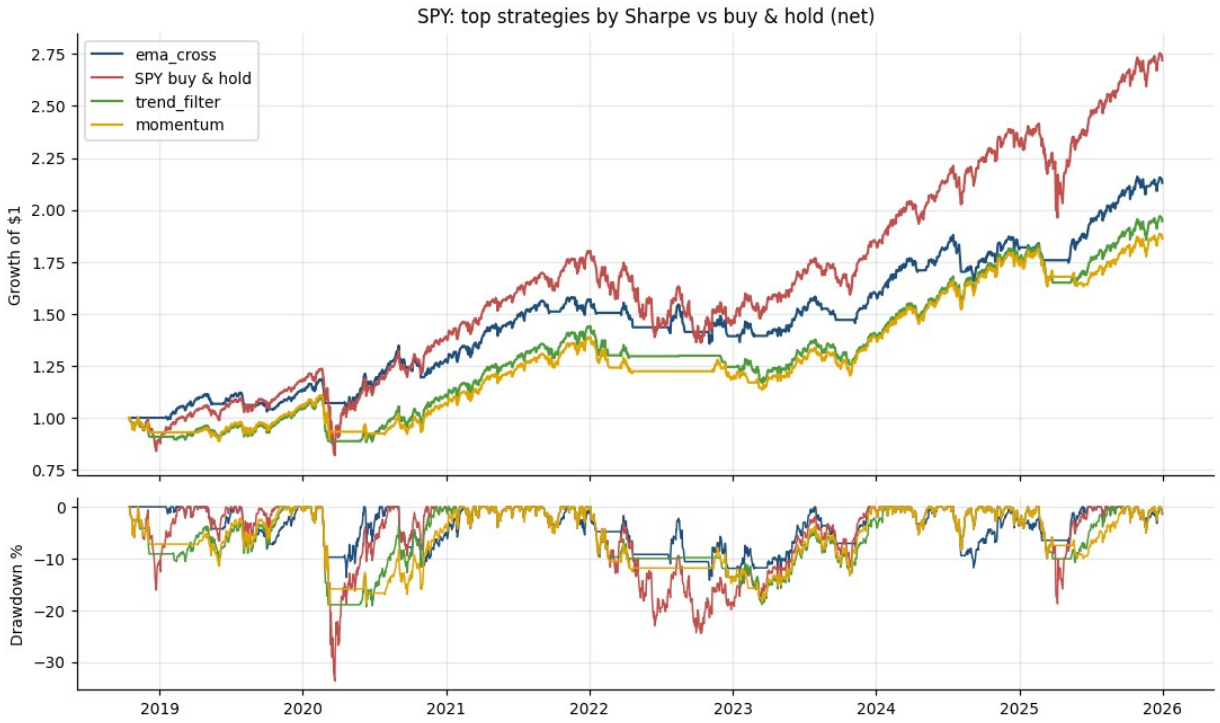
AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/evaluate.py add an 'Ulcer index' column = sqrt(mean(drawdown% squared)) computed from common.metrics.drawdown_series of each strategy's returns, without changing common/metrics.py. Show it in the printed table and the CSV, and explain in one sentence what it adds to max drawdown. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: The new column shows a much lower Ulcer index for ema_cross than for SPY buy & hold.

Expected Output (Full Code)

python -m pipeline evaluate --symbol SPY: window 2018-10-16 to 2025-12-31; ema_cross CAGR 11.09%, Sharpe 0.95, max drawdown -14.19% (best Sharpe); SPY buy & hold 15.28%, 0.81, -33.72%; trend_filter 9.70%, 0.81, -19.36%, 24 trades, win rate 41.67%, profit factor 3.79, beta 0.38; bollinger_breakout last (Sharpe 0.35); files _outputs/Lab09_metrics.csv and chart _outputs/Lab09_metrics.png.

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Lab 9 output (labs_fullcode/_outputs/Lab09_metrics.png), offline mode on the bundled data.

Debrief

Performance metrics (K3) answer different questions: CAGR measures growth, Sharpe and Sortino reward per unit of risk, drawdown and Calmar the pain an investor must survive, beta how much is just the market. Running every strategy through one evaluation stage (A3) is what makes the comparison fair.

Troubleshooting

Symptom	What to do
Sharpe looks too high	Check that daily (not cumulative) returns are used and annualised with sqrt(252).
Win rate below 50% yet profitable	Expected for trend following: few large winners pay for many small losers (see profit factor).
Strategies cover different periods	Slice every return series from the same start date (first valid 200-day SMA).

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